

Vedant Pol

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SKILLS

Languages: Python, SQL, Java, Bash

GenAI & LLMs: Retrieval-Augmented Generation, LangChain, LangGraph, LLM Evaluation, Prompt Engineering, LLM-as-Judge, Agentic Workflows, Guardrails, Prompt Injection Testing

Retrieval & Vector Search: Embeddings, Pinecone, Milvus, FAISS, Semantic Search, Hybrid Search, Reranking, Metadata Filtering, Chunking Strategies, Citation-Grounded Generation

ML, MLOps & Tools: PyTorch, Hugging Face Transformers, scikit-learn, MLflow, Docker, Kubernetes, AWS SageMaker, Kafka, Spark, Playwright, Pytest, Allure, GoCD

WORK EXPERIENCE

IDFC First Bank, Mumbai

2024 - Present

Software Development Engineer in Test, AI Systems & Automation

- Built and evaluated a GenAI-based fraud-risk analysis workflow combining rule-based checks, anomaly signals, vector retrieval, and LLM-generated investigation summaries across **284K+ transaction records**, improving fraud-case triage recall by **25%**.
- Developed a Retrieval-Augmented Generation system for compliance and policy search using **LangChain, Pinecone, embeddings, metadata filtering, and citation-grounded retrieval**; indexed **500+ regulatory and operational documents**, reducing manual policy lookup time by **60%**.
- Designed a **RAG benchmarking framework** to compare chunking strategies, embedding models, top-k settings, reranking, and prompt templates; improved grounded-answer accuracy from **78% to 91%**.
- Created reusable LLM and RAG evaluation datasets with golden questions, expected source documents, citation checks, adversarial prompts, and missing-context queries; tracked **Recall@K, Precision@K, MRR, faithfulness, hallucination rate, latency, and cost per query**.
- Built an automated LLM evaluation pipeline using **Playwright, Python, GoCD, and Allure** to validate factuality, refusal behavior, hallucination, prompt-injection resistance, and response consistency, reducing manual review cycles by **50%**.
- Python, SQL, LangChain, Pinecone, RAG, LLM Evaluation, Prompt Engineering, Playwright, GoCD, Allure, MLflow, Docker, Kubernetes, Kafka, Spark

Google Summer of Code, XWiki

Jun 2022 - Sep 2022

Software Engineering Intern, Open Source

Remote

- Built the **XWiki Snap package** and automated cloud-ready release workflows using CI/CD, Linux packaging, Docker, Kubernetes, and deployment automation.
- Contributed **24 PRs, 85 commits, and 7K+ lines of code** (Commits, Report); created user docs and developer docs for setup, builds, packaging, automation, and deployment.
- GitHub Actions, Linux Packaging, CI/CD, Docker, Kubernetes, Bash, Open Source, Developer Documentation

EDUCATION

Thakur College of Engineering & Technology

2020 - 2024

Bachelor of Engineering in Computer Science

CGPA: 9.08/10

Relevant Coursework: Data Structures and Algorithms, Databases, Operating Systems, Computer Networks, Machine Learning, Artificial Intelligence, Software Engineering, Cloud Computing

PROJECT WORK

- **Enterprise RAG Knowledge Assistant:** Built a GenAI assistant for policy and technical document search using semantic retrieval, metadata filtering, reranking, and citation-grounded answer generation. Implemented PDF parsing, text cleaning, chunking, embedding generation, vector indexing, and source-aware retrieval. Python, LangChain, Pinecone, FastAPI, Docker.
- **Multi-Agent Knowledge Management System:** Developed an agentic LLM workflow using LangGraph with Retriever, Planner, Critic, and Summarizer agents. Implemented guardrails for prompt injection, unsupported claims, citation validation, and hallucination detection using rule-based checks and LLM-as-Judge evaluation. LangGraph, AutoGen, FastAPI, Docker.
- **LLM Evaluation and Regression Testing Framework:** Built an automated framework to evaluate factuality, safety, hallucination rate, formatting quality, latency, deterministic business-rule compliance, citation presence, retrieval quality, adversarial prompts, and missing-context handling. Python, Playwright, Pytest, Allure, MLflow.

PATENT

- **ML-Based Dot Matrix Printer Fault Diagnosis** (German Patent): Developed an audio-based supervised ML system using Python, signal processing, feature extraction, classification models, and hardware sensor data to detect paper jams, missing print lines, carriage faults, and print head issues. Python, Machine Learning, Signal Processing, CNNs, Raspberry Pi.

AWARDS AND CERTIFICATES

- **AWS Certified Machine Learning – Specialty:** Credential
- **AWS Certified Cloud Practitioner:** Credential
- **Mathematics for Machine Learning, Imperial College London:** Certificate